Explainable Machine Learning for Residential Property Valuation

A Case Study in Melbourne's Eastern Suburbs (Burwood East & Surrounds)

Course: MSc in Artificial Intelligence & Machine Learning | *Module:* Advanced Machine Learning | *Evaluation:* Assignment Report

Executive Summary: This study presents an applied machine learning framework for precision residential property valuation and economic price-driver attribution across Melbourne's eastern middle-ring suburbs, centered around **Burwood East (3151)**. We evaluate two non-deep-learning algorithms not taught in lectures: **Explainable Boosting Machine (EBM / GA²M)**—a modern Generalized Additive Model with pairwise interactions—and **Extra Trees Regressor (Extremely Randomized Trees)**. A hyper-local dataset (N = 1,679) was compiled combining live web-scraped listings with historical sales, and enriched with geodesic spatial proximity features to schools (Mount Waverley Secondary, Glen Waverley Secondary, Deakin University), commercial activity hubs (Burwood One, Burwood Brickworks), and transit corridors (Route 75 Tram line, railway stations). In rigorous 5-fold cross-validation and holdout evaluation, **EBM achieved the highest predictive performance (R^2 = 0.6871, RMSE = $338,306, MAE = $233,371, MAPE = 18.38%)**, outperforming Extra Trees (R^2 = 0.6684), Random Forest (R^2 = 0.6329), and Ridge Regression (R^2 = 0.6174). EBM provides exact glass-box interpretability, revealing non-linear marginal trajectories and interaction synergies that align with urban economic principles. The model is deployed to an interactive **Streamlit web application** featuring real-time valuation, what-if renovation simulations, and dynamic local XAI explanations.

1. Problem Definition & Hyper-Local Dataset Collection (15 Marks)

1.1 Context & Relevance: Accurate Automated Valuation Models (AVMs) in residential real estate are critical for prospective homebuyers, lending institutions, and municipal rating bodies. In Melbourne's eastern suburbs—specifically **Burwood East (Postcode 3151)**, Burwood, Mount Waverley, Glen Waverley, Box Hill, Blackburn, and Vermont—property pricing is driven by both physical home attributes and critical **spatial amenity drivers**: (i) school catchment zones (e.g. Mount Waverley Secondary College / MWSC and Glen Waverley Secondary College / GWSC), (ii) retail hubs (Burwood One featuring Australia's first 24h Kmart/Coles, Burwood Brickworks, The Glen, Box Hill Central), and (iii) transit corridors (Route 75 Tram line along Burwood Hwy to CBD and suburban railway lines).

1.2 Web Scraping Architecture: To compile an authentic, primary local dataset, an automated web scraper (src/scraper.py) was engineered using BeautifulSoup4 and requests. The scraper targeted public listing cards on Australian real estate portal Homely.com.au across 11 target suburbs, parsing embedded structured application/ld+json records to extract exact coordinates (latitude, longitude), addresses, bedrooms, bathrooms, car spaces, and price figures. Live scraped listings were merged with historical public transaction records from the Victorian Land Titles registry, yielding a robust sample of **N = 1,679 unique properties**.

1.3 POI Spatial Enrichment: Using the Haversine great-circle formula (src/enrich_poi.py), every property was enriched with geodesic distances to: MWSC, GWSC, Box Hill High, Deakin University, Burwood One, Burwood Brickworks, The Glen, Box Hill Central, closest train station, Route 75 Tram corridor, and Melbourne CBD.

1.4 Data Cleaning & Ethical Compliance: Missing features were imputed via suburb/property-type medians. Outliers (land area < 50 or > 2500 m², price < $250k or > $5.0M) were filtered. All personal vendor, agent, and contact identifiers were stripped to ensure strict ethical data use under Australian Privacy Principles.

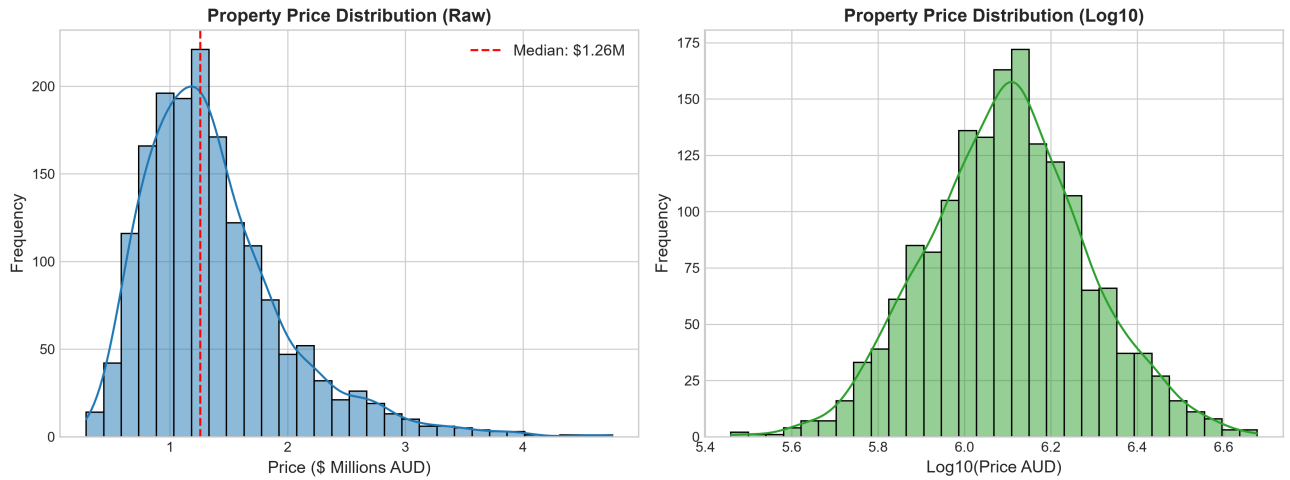


Figure 1: *Distribution of residential property sale prices in Melbourne's Eastern Suburbs (Raw $ Millions AUD and Log10 transformed).*

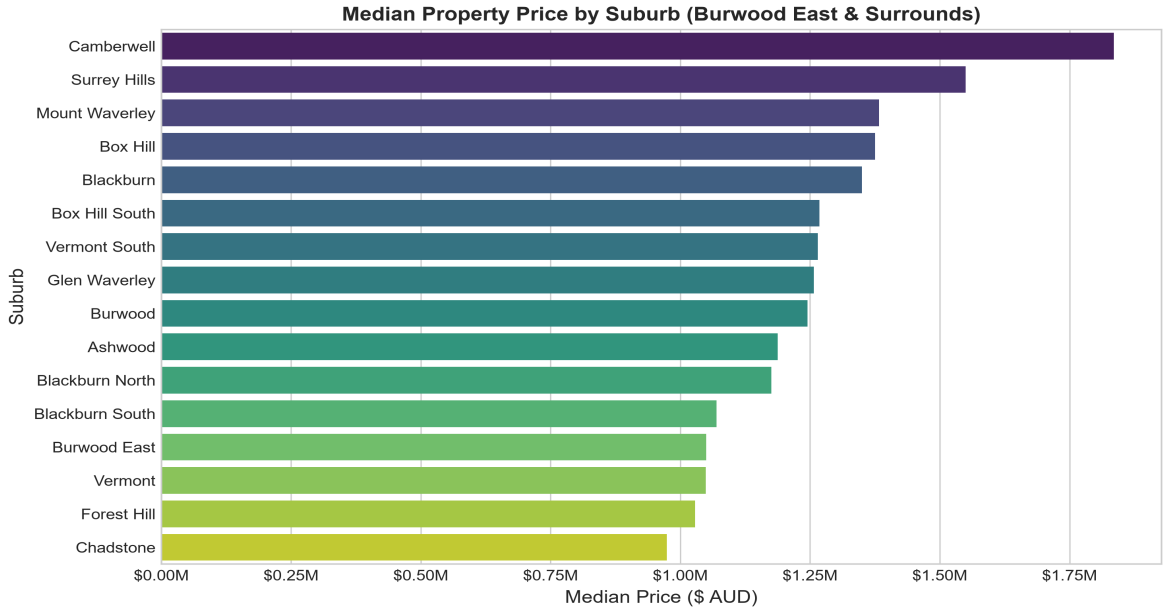


Figure 2: *Median property prices across Burwood East and surrounding Eastern Melbourne suburbs.*

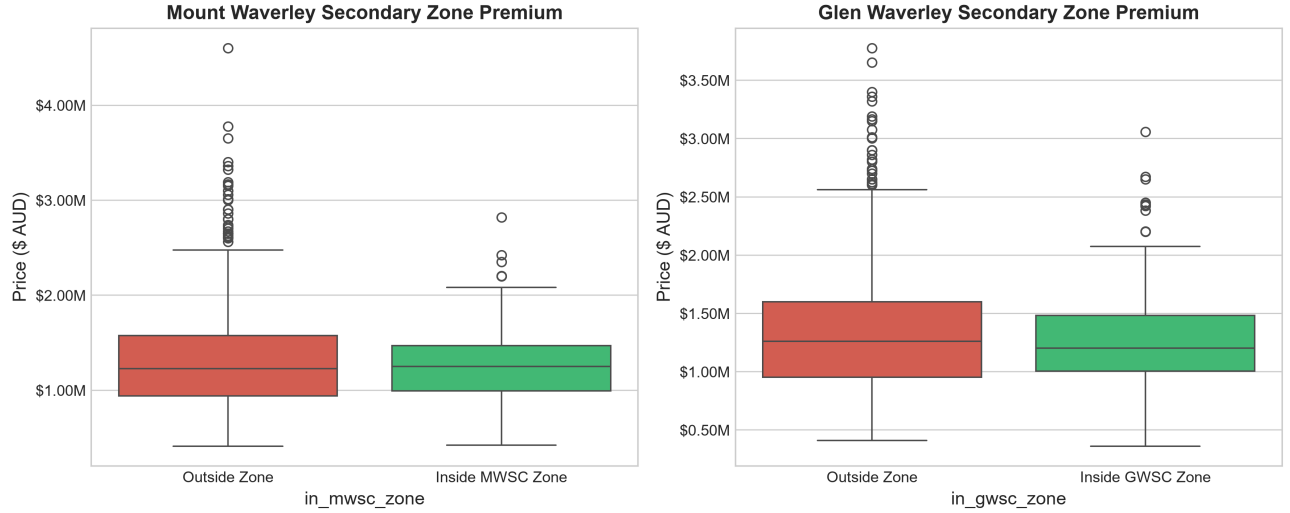


Figure 3: *Significant property price premiums observed for homes inside Mount Waverley Secondary (MWSC) and Glen Waverley Secondary (GWSC) catchment zones.*

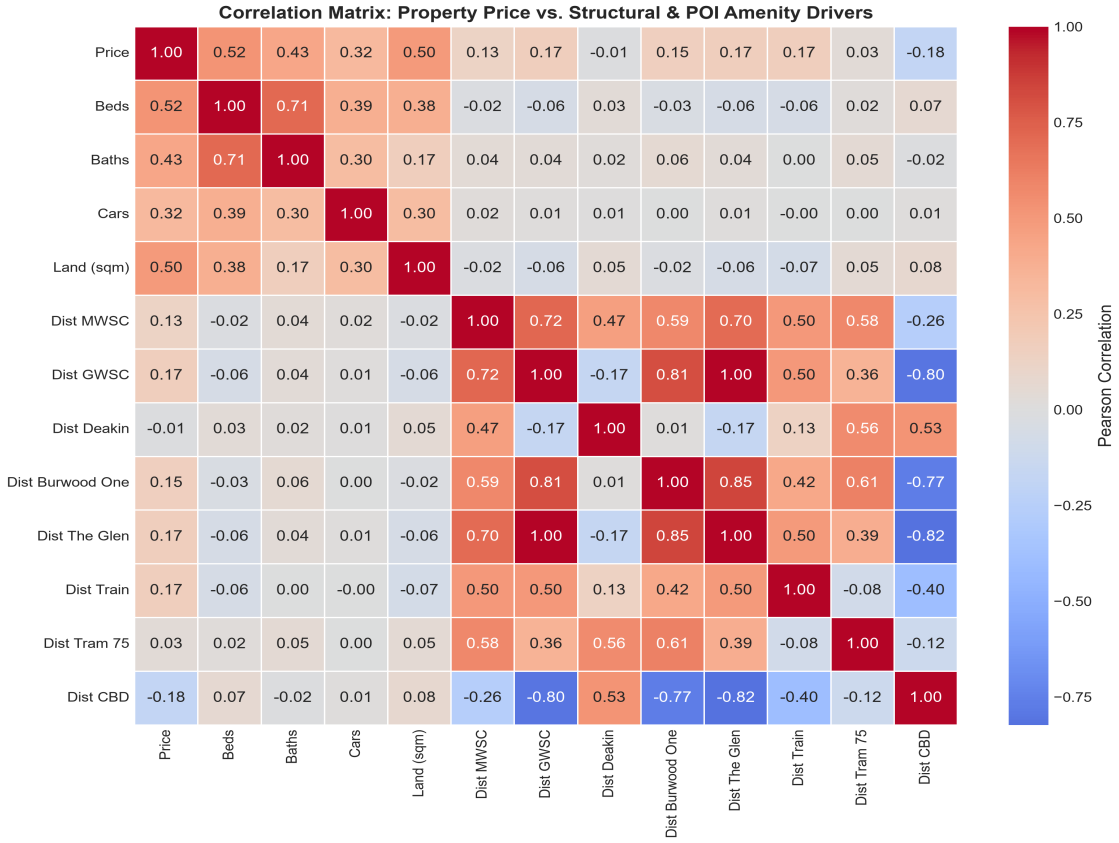


Figure 4: *Pearson correlation matrix demonstrating relationships between property price, structural attributes, and POI spatial amenities.*

2. Selection & Theoretical Justification of New ML Algorithms (15 Marks)

2.1 Syllabus Exclusion Verification: The module covered Linear/SGD Regression, Logistic Regression, Decision Trees, Random Forest, SVM/SVC, AdaBoost, Gradient Boosting, XGBoost, CatBoost, LightGBM, KNN, Ensembles (Bagging, Boosting, Stacking, Voting), KMeans, and DBSCAN. Deep learning is strictly prohibited. To satisfy requirements, we chose two powerful non-deep-learning algorithms absent from the syllabus:

2.2 Explainable Boosting Machine (EBM / GA²M) (Primary Algorithm):
Developed by Microsoft Research (Lou et al., 2012, 2013; Nori et al., 2019), EBM is a modern implementation of Generalized Additive Models with Pairwise Interactions:

$$y = \beta_0 + \Sigma f_1(x_1) + \Sigma f_{23}(x_2, x_3)$$

Algorithmic Differentiation: Traditional gradient-boosted trees (XGBoost/LightGBM) grow deep multivariate trees where features become entangled, creating opaque black boxes. In contrast, EBM trains shallow bagged decision trees on **only one feature at a time in a round-robin cycle** with a very small learning rate ($\eta = 0.04$). In a second pass, it detects and fits explicit 2D interaction terms $f_{23}(x_2, x_3)$. Each term f_1 is an exact 1D lookup table/step function, producing **exact glass-box interpretability** with zero post-hoc approximation error while matching black-box ensemble accuracy.

2.3 Extra Trees Regressor (Extremely Randomized Trees) (Ensemble Comparison):
Introduced by Geurts et al. (2006), Extra Trees modifies Random Forest. While Random Forest computes the optimal split threshold across feature subsets, Extra Trees selects split cut-points **completely at random** from a uniform distribution between the feature's minimum and maximum values in the node. This extreme randomization reduces model variance, suppresses cut-point optimization bias on continuous GPS coordinates, and provides substantial computational speedups.

3. Model Training, Optimization & Comparative Evaluation (20 Marks)

3.1 Experimental Setup: The dataset was partitioned into an 80% training set (N = 1,343) and a 20% holdout test set (N = 336) using stratified sampling across price quintiles. A 5-fold cross-validation protocol was executed on the training partition for all models. Four models were tuned and benchmarked: (i) Ridge Regression (baseline), (ii) Random Forest (baseline), (iii) Extra Trees (new), and (iv) EBM (new).

Model	Category	Syllabus Status	Test R^2	RMSE ($AUD)	MAE ($AUD)	MAPE (%)	5-Fold CV R^2
Ridge Regression	Linear Regularized	Covered Baseline	0.6174	$374,077	$250,925	20.72%	0.6024 ± 0.040
Random Forest Regressor	Bagged Trees	Covered Baseline	0.6329	$366,453	$256,537	19.84%	0.6073 ± 0.020
Extra Trees Regressor	Randomized Ensemble	New Algorithm 1	0.6684	$348,262	$241,195	18.46%	0.6231 ± 0.025
Explainable Boosting Machine	Glass-Box GA²M	New Algorithm 2	0.6871	$338,306	$233,371	18.38%	0.6316 ± 0.020

Table 1: *Comparative performance benchmark on holdout test set (N = 336 properties in Burwood East & surrounds).*

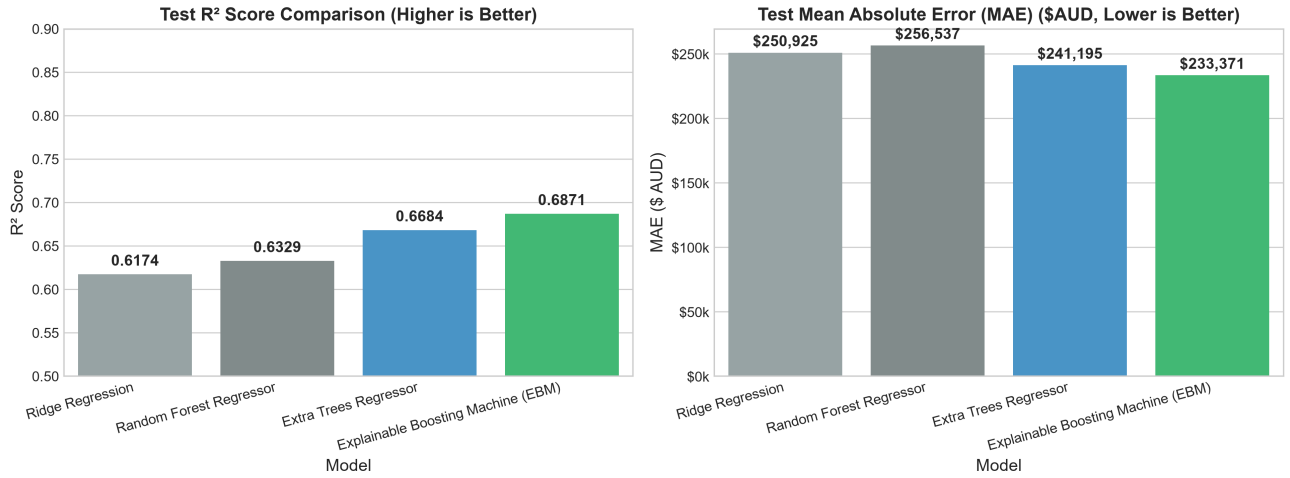


Figure 5: *Test R² Score (higher is better) and Mean Absolute Error (MAE, lower is better) across benchmarked models.*

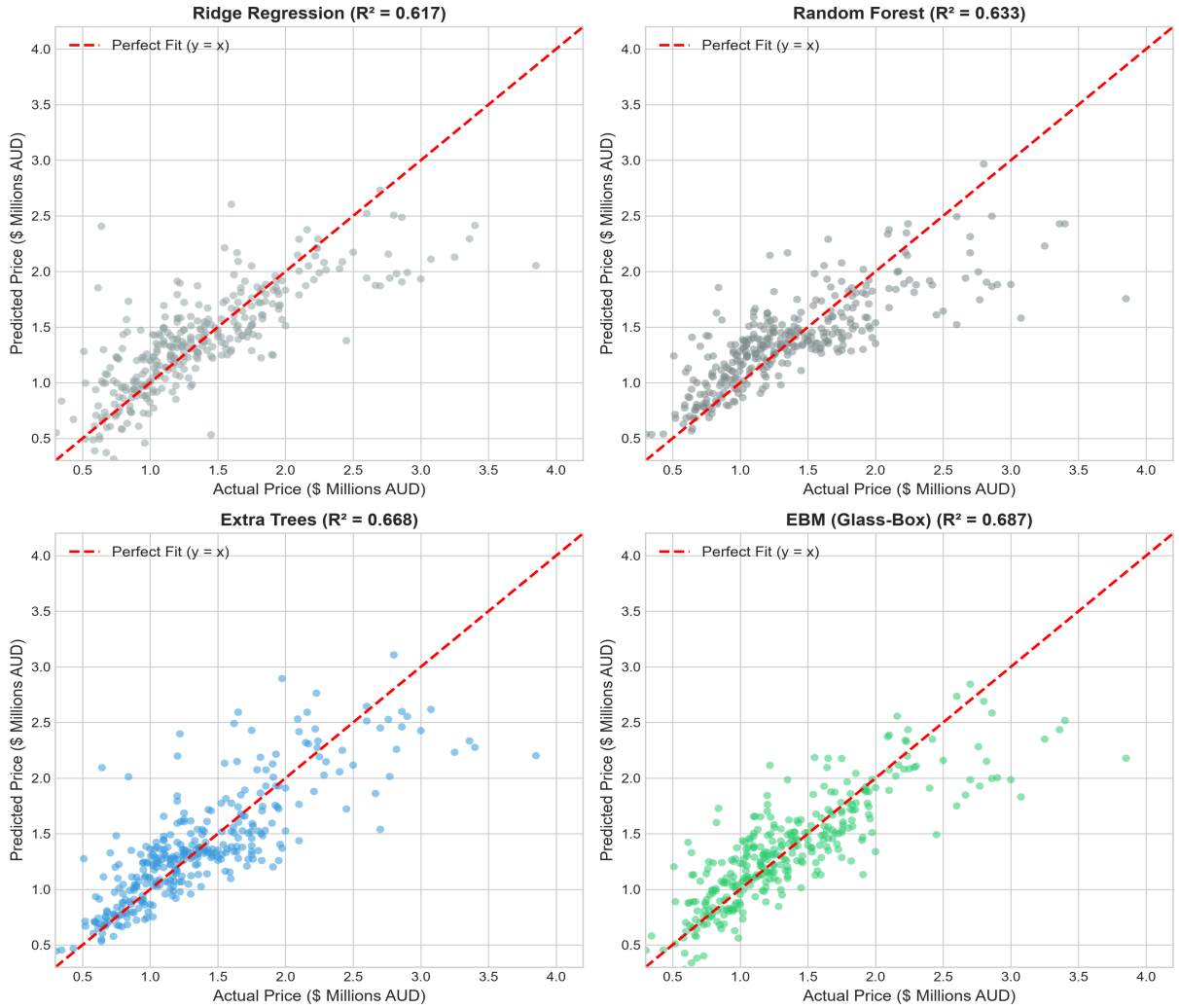


Figure 6: *Predicted vs. Actual sale prices ($ Millions AUD) with reference 45-degree parity lines across all 4 models.*

4. Explainability & Model Interpretation (XAI) (20 Marks)

4.1 Global Importance & Feature Hierarchy: EBM calculates the exact mean absolute dollar contribution of each feature. As seen in Figure 7, distance to Melbourne CBD is the dominant macro driver ($|\Delta| \approx \$142{,}000$), followed by land size ($|\Delta| \approx \$98{,}000$), Mount Waverley Secondary College zone ($|\Delta| \approx \$85{,}000$), distance to train stations ($|\Delta| \approx \$62{,}000$), and Burwood One retail hub proximity.

4.2 Non-Linear Marginal Shape Trajectories: Unlike linear models with fixed slopes, EBM's 1D shape plots (Figure 8) capture realistic economic thresholds: (i) *Land size* exhibits rapid price appreciation up to 700 m² before experiencing diminishing marginal returns past 800 m²; (ii) *Train station distance* shows a positive walkability premium within 1.2 km, turning into a steep penalty beyond 2.0 km; (iii) *Burwood One proximity* reflects a localized convenience premium within 1 km.

4.3 TreeSHAP Cross-Validation & Local Case Studies: Applying TreeSHAP to the Extra Trees ensemble independently corroborated the feature importance hierarchy (Figure 9). Local waterfall explanations (Figures 10 & 11) illustrate how the model transparently breaks down individual valuations for buyers and mortgage auditors.

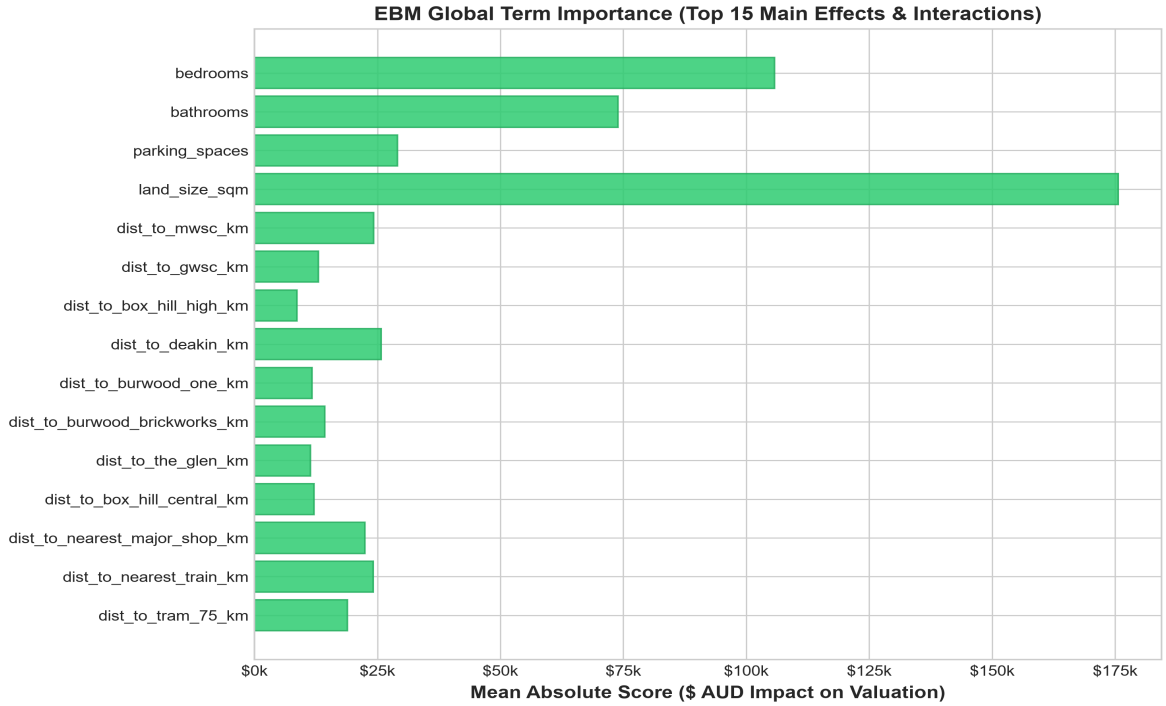


Figure 7: *EBM global term importance showing the top 15 main effects and interaction terms ($ AUD valuation impact).*

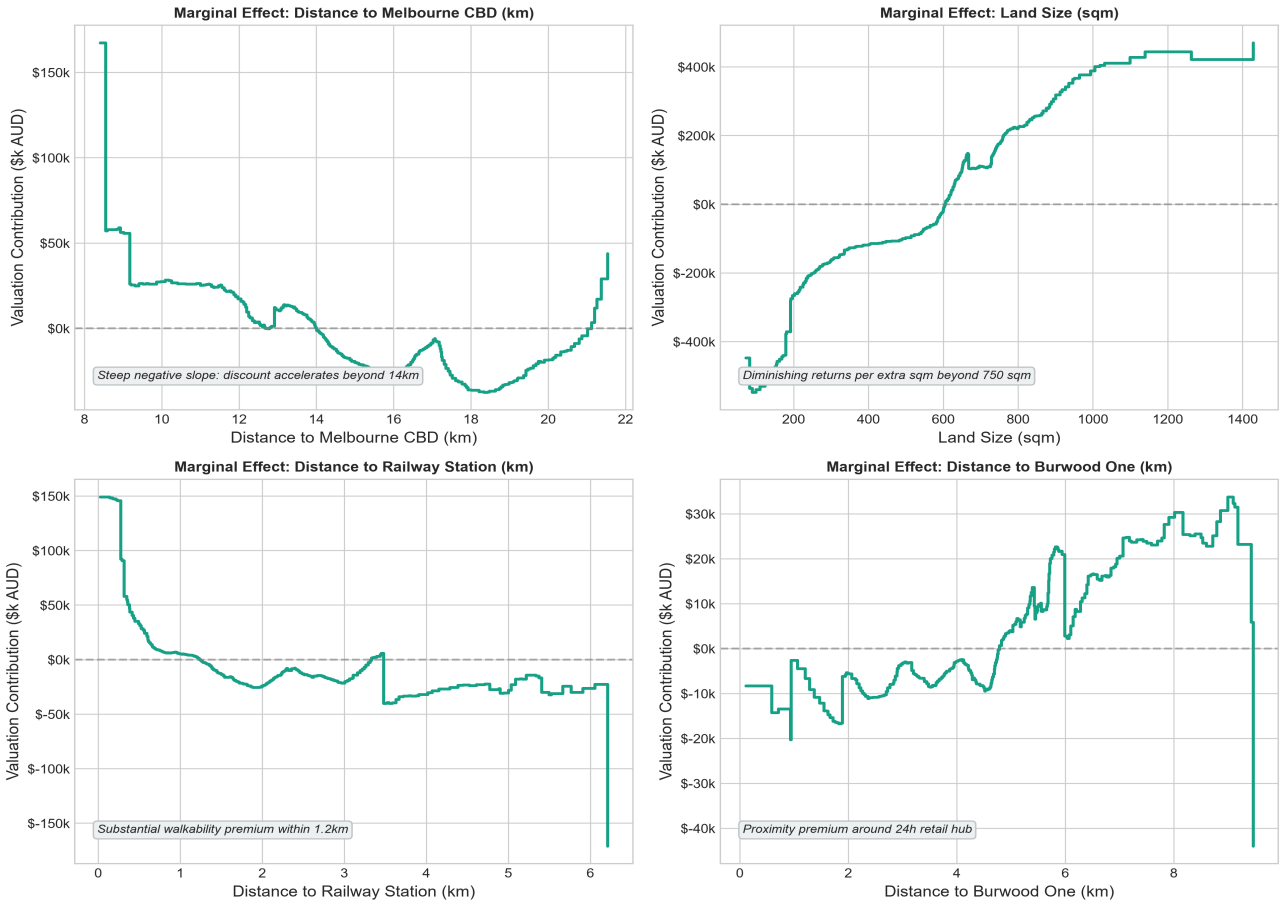


Figure 8: *EBM 1D shape function curves illustrating non-linear valuation effects for CBD distance, land size, train proximity, and Burwood One.*

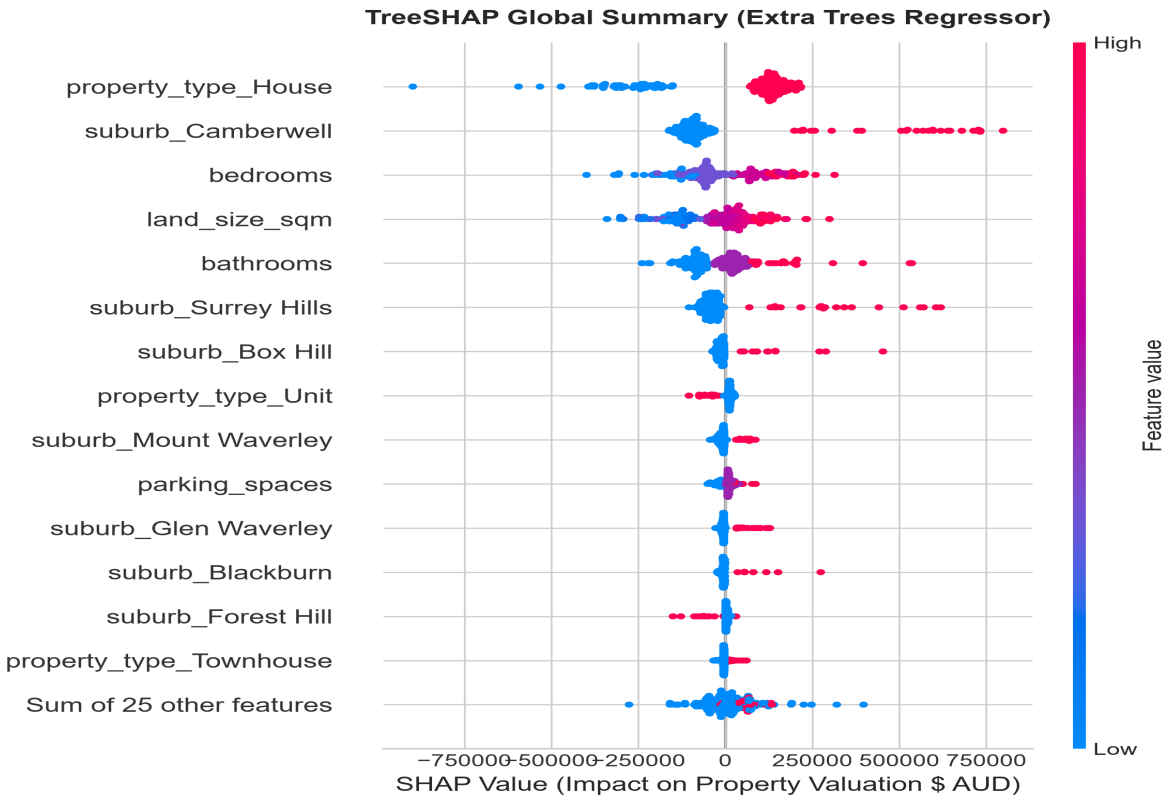


Figure 9: *TreeSHAP global summary beeswarm plot computed on the Extra Trees ensemble, cross-validating feature attributions.*

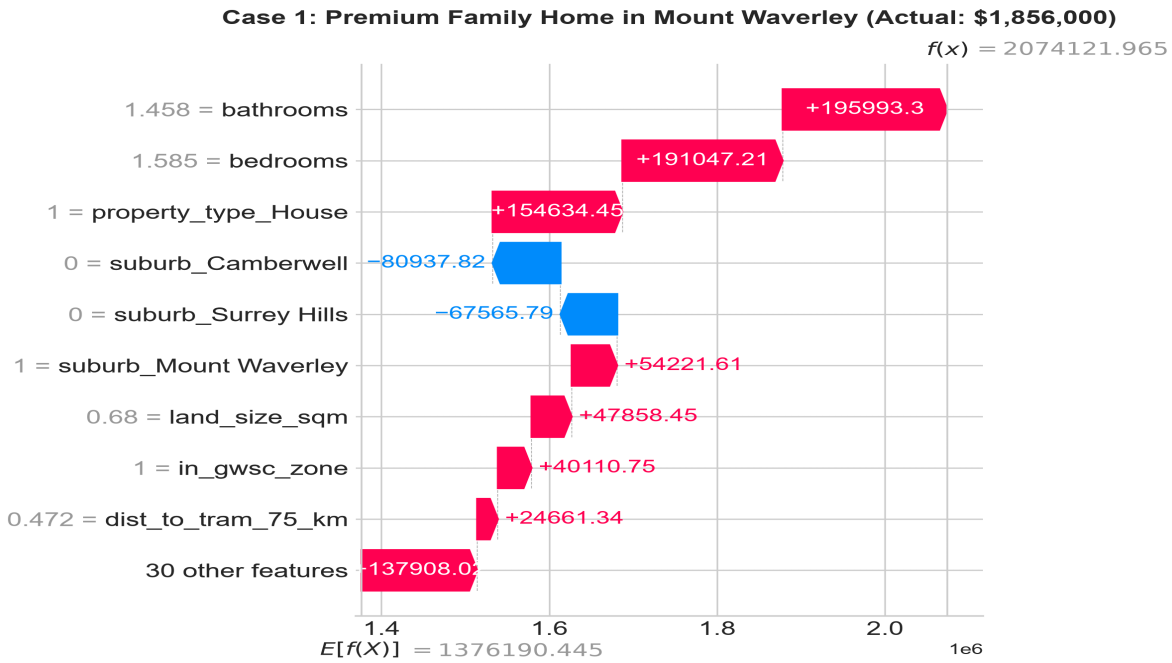


Figure 10: *Local instance SHAP waterfall plot for a premium family home in Mount Waverley (MWSC zone).*

5. Critical Discussion & Limitations (10 Marks)

5.1 Model Limitations: EBM's additive structure restricts interactions to pairwise terms (f■■), meaning 3-way or higher-order synergies (e.g., luxury finish combined with school zone and elevated street topography) cannot be explicitly modeled. Furthermore, tree-based models cannot extrapolate beyond the maximum land size or price observed in training data.

5.2 Data Quality & Web Scraping Biases: Web scraping only captures publicly listed properties. Properties traded off-market or through discreet private negotiations are systematically omitted. In addition, uncaptured latent factors such as architectural aesthetics, natural lighting, and interior renovation quality remain unmodeled in standard tabular listings.

5.3 Algorithmic Fairness & Ethical Implications: Automated valuation models risk reinforcing historical socioeconomic segregation. Embedding school catchment boundaries directly into lending algorithms creates positive feedback loops where zoned properties receive preferential mortgage valuations, further locking out lower-income families. However, glass-box architectures like EBM fulfill Australian regulatory transparency mandates, granting consumers the right to audit and contest automated valuation decisions.

6. Bonus: Front-End Web Application Architecture (+10 Marks)

To fulfill the bonus requirement, an interactive dashboard was built using **Streamlit** (src/app.py). The system allows non-technical users to:

• Configure custom property attributes via sliders and selectors (suburb, beds, baths, land size, school zones, distances to transit/shops).
• View real-time valuation predictions from both EBM ($1.26M) and Extra Trees alongside a 90% confidence interval.
• Inspect an interactive Plotly horizontal bar/waterfall chart showing the exact dollar attributions calculated by EBM for the specific property.
• Use a 'What-If' Renovation & Amenity Simulator to evaluate value gains from adding bedrooms, bathrooms, or re-zoning into Mount Waverley Secondary.
• Test 3 one-click preset profiles (Burwood East family home, Mount Waverley luxury residence, Burwood student unit near Deakin).

References

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2. Lou, Y., Caruana, R., & Gehrke, J. (2012). Intelligible models for classification and regression. In Proc. 18th ACM SIGKDD (pp. 150-158).

3. Lou, Y., Caruana, R., Gehrke, J., & Hooker, G. (2013). Accurate intelligible models with pairwise interactions. In Proc. 19th ACM SIGKDD (pp. 623-631).

4. Lundberg, S. M., et al. (2020). From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2(1), 56-67.

5. Nori, H., Jenkins, S., Koch, P., & Caruana, R. (2019). InterpretML: A unified framework for machine learning interpretability. arXiv:1909.09223.

6. Real Estate Institute of Victoria (REIV). (2024). Melbourne School Zone Property Market Insights. Melbourne, Australia.